Squashed entanglement at the Bell edge and a fidelity–Holevo deficit bound

Abstract

For the two-qubit state $\rho_c = (|00\rangle\langle 00| + |11\rangle\langle 11| + c|00\rangle\langle 11| + c|11\rangle\langle 00|)/2$, the infimum of half the conditional mutual information over all finite quantum conditioning systems satisfies

$$\frac{c^2}{2\ln 2}\ln\frac{1}{4c} \le E_{\mathrm{sq}}(\rho_c) \le h_2\left(\frac{1+\sqrt{1-c^2}}{2}\right) \quad (0 < c \le \tfrac{1}{4}).$$

Thus $E_{\mathrm{sq}}(\rho_c) = (c^2/2)\log_2(1/c) + O(c^2)$. The proof uses an exact reduction to finite density pairs, a jointly convex column entropy deficit, and a common-channel inequality bounding the square of the increase in root fidelity by the binary Holevo loss. All matrix statements include singular and noncommuting densities.

The upper bound is the entanglement-of-formation formula of Wootters [3], combined with $E_{\mathrm{sq}} \le E_F$ [4]. The contribution here is the lower bound with the same leading behavior as $c \downarrow 0$. For comparison, the hashing inequality [5] and $E_{\mathrm{sq}} \ge E_D$ [4] also give

$$E_{\mathrm{sq}}(\rho_c) \ge 1 - h_2\left(\frac{1+c}{2}\right).$$

Indeed ρ_c has maximally mixed marginals and nonzero eigenvalues $(1 \pm c)/2$. This bound can be stronger away from the asymptotic regime (for example at $c = 0.1$), so the new lower bound is not claimed to dominate it throughout the interval.

1 Definitions and statements

All Hilbert spaces in this paper are finite dimensional and complex. A density matrix is positive semidefinite with trace one. A channel is a completely positive trace-preserving linear map. Write

$$S(\rho) = -\operatorname{Tr}\rho\log_2\rho, \qquad I(A:B\mid E)_\omega = S(AE) + S(BE) - S(E) - S(ABE),$$

where $0\log 0 = 0$. Squashed entanglement means

$$E_{\mathrm{sq}}(\rho_{AB}) = \inf_{\substack{\dim E<\infty,\ \omega_{ABE}\ \text{density} \\ \operatorname{Tr}_E\omega=\rho_{AB}}} \frac{1}{2} I(A:B\mid E)_\omega. \tag{1.1}$$

The conditioning dimension is included in the infimum. For two densities on a common space define their affinity, root fidelity, equal-prior Holevo information, and cost by

$$a(\rho,\sigma) = \operatorname{Tr}\sqrt{\rho}\sqrt{\sigma}, \qquad F(\rho,\sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1, \tag{1.2}$$

$$\chi(\rho,\sigma) = S\left(\frac{\rho+\sigma}{2}\right) - \frac{S(\rho)+S(\sigma)}{2}, \qquad V(\rho,\sigma) = 1 - \chi(\rho,\sigma). \tag{1.3}$$

The trace in the affinity is real and nonnegative. Cauchy–Schwarz and the trace norm give $0 \le a \le F \le 1$. Let

$$h_2(t) = -t\log_2 t - (1-t)\log_2(1-t), \qquad \phi(v) = h_2\left(\frac{1+\sqrt{1-v}}{2}\right), \quad 0 \le v \le 1.$$

Theorem 1.1. *For every* $0 < c \le 1/4$,

$$\frac{c^2}{2\ln 2}\ln\frac{1}{4c} \le E_{\mathrm{sq}}(\rho_c) \le \phi(c^2). \tag{1.4}$$

There exist constants $C, c_0 > 0$ *such that, for* $0 < c \le c_0$,

$$\left| E_{\mathrm{sq}}(\rho_c) - \frac{c^2}{2}\log_2\frac{1}{c} \right| \le Cc^2, \qquad 0 \le \phi(c^2) - E_{\mathrm{sq}}(\rho_c) \le Cc^2.$$

Theorem 1.2. *For any finite density pair* η_0, η_1 *and any common channel* $\mathcal{N}$*, including singular inputs,*

$$\left[F(\mathcal{N}\eta_0, \mathcal{N}\eta_1) - F(\eta_0, \eta_1)\right]^2 \le 4\ln 2\,\left[\chi(\eta_0, \eta_1) - \chi(\mathcal{N}\eta_0, \mathcal{N}\eta_1)\right]. \tag{1.5}$$

Theorem 1.3. *For every finite density pair* ρ, σ *of affinity* $c > 0$*, with* $F = F(\rho, \sigma)$,

$$1 - \chi(\rho, \sigma) \ge \phi(F^2) - \phi(F^2 - c^2) + \frac{(F - \sqrt{Fc})^2}{4\ln 2}. \tag{1.6}$$

2 Exact reduction to density pairs

Proposition 2.1. *For* $0 \le c \le 1$,

$$E_{\mathrm{sq}}(\rho_c) = \inf_{\substack{\rho,\sigma \text{ finite densities} \\ a(\rho,\sigma)=c}} V(\rho, \sigma). \tag{2.1}$$

Both sides range over every finite dimension.

Proof. Given a finite pair with affinity c, place the following block matrix on $\mathrm{span}\{|00\rangle, |11\rangle\} \otimes E$:

$$\Gamma = \frac{1}{2}\begin{pmatrix} \rho & \sqrt{\rho}\sqrt{\sigma} \\ \sqrt{\sigma}\sqrt{\rho} & \sigma \end{pmatrix}. \tag{2.2}$$

It is a column Gram, has trace one, and has AB marginal ρ_c. Both AE and BE marginals are the block diagonal state $\frac{1}{2}(\rho \oplus \sigma)$. The E marginal is $(\rho+\sigma)/2$. If $M = (\sqrt{\rho}, \sqrt{\sigma})^{\mathsf{T}}/\sqrt{2}$, then $\Gamma = MM^\dagger$ and $M^\dagger M = (\rho + \sigma)/2$. The opposite rectangular Grams have the same nonzero eigenvalues. Consequently

$$\frac{1}{2}I(A : B \mid E)_\Gamma = 1 + \frac{S(\rho) + S(\sigma)}{2} - S\left(\frac{\rho+\sigma}{2}\right) = V(\rho, \sigma).$$

This proves the inequality from the left side of (2.1) to the right.

Conversely let ω_{ABE} be any extension in (1.1). Positivity confines its support to $\mathrm{span}\{|00\rangle, |11\rangle\} \otimes E$. Indeed an outside-support diagonal entry of its AB marginal is zero; the nonnegative diagonal entries over E must all vanish, and a zero diagonal of a positive matrix forces its corresponding row and column to vanish. A finite purification therefore has the form

$$|\Psi\rangle = \frac{|00\rangle|w_0\rangle + |11\rangle|w_1\rangle}{\sqrt{2}}, \qquad \|w_i\| = 1, \quad \langle w_1, w_0\rangle = c,$$

with $w_i \in E \otimes G$. Let ρ_i and τ_i be the E and G marginals of $|w_i\rangle$, and let their Holevo quantities be χ_E, χ_G. The conditional marginal spectra of a pure vector agree. The block entropy formula and purity of Ψ give

$$J := \frac{1}{2}I(A : B \mid E)_\omega = 1 - \frac{\chi_E + \chi_G}{2} \ge 0. \tag{2.3}$$

Reshape w_i into rectangular matrices M_i of Hilbert–Schmidt norm one. For any Hermitian X, Y, spectral decomposition gives

$$\begin{aligned}\||X| - |Y|\|_{\mathrm{HS}}^2 &= \sum_{j,k} (|\lambda_j| - |\mu_k|)^2 |\langle j,k\rangle|^2 \\ &\le \sum_{j,k} (\lambda_j - \mu_k)^2 |\langle j,k\rangle|^2 = \|X - Y\|_{\mathrm{HS}}^2.\end{aligned}$$

Apply this to $X_i = \left(\begin{smallmatrix} 0 & M_i \\ M_i^\dagger & 0 \end{smallmatrix}\right)$. Its absolute value has blocks $\sqrt{M_i M_i^\dagger}$ and $\sqrt{M_i^\dagger M_i}$. If a_E, a_G are the affinities of the two marginal pairs, this yields

$$4 - 2(a_E + a_G) \le 4 - 4c, \qquad a := \frac{a_E + a_G}{2} \ge c.$$

The possible transpose in the vectorized G marginal preserves this affinity. The equal direct sum $\widehat{\rho}_i = (\rho_i \oplus \tau_i)/2$ has affinity a and cost J; the common flag entropy cancels. If $c > 0$, mix this pair in an orthogonal block, with weight c/a, with two mutually orthogonal pure states. The latter pair has affinity zero and cost zero. The new finite pair has affinity exactly c and cost $(c/a)J \le J$. Thus every finite extension produces a finite admissible pair of no larger cost. Taking infima proves the reverse inequality. At $c = 0$, ρ_0 has a labelled product extension of zero CMI, and the orthogonal pure pair has zero cost, so both infima are zero. □

3 Common recovery and the fidelity–Holevo inequality

For densities α, β write

$$D_2(\alpha\|\beta) = \operatorname{Tr} \alpha(\log_2 \alpha - \log_2 \beta)$$

with value $+\infty$ when the support inclusion fails. $D_{M,2}$ is the supremum of classical relative entropy in bits over all finite POVMs. We use the common-reference recovery theorem [1, Corollary 4.2, equations (60)–(61)]: if α is supported in a density ζ, then

$$D_2(\alpha\|\zeta) - D_2(\mathcal{N}\alpha\|\mathcal{N}\zeta) \ge D_{M,2}(\alpha\|\mathcal{R}_{\zeta,\mathcal{N}}\mathcal{N}\alpha). \tag{3.1}$$

The completely positive recovery map depends only on $\zeta, \mathcal{N}$. With $\mu = \mathcal{N}\zeta$, it is the average of

$$\mathcal{R}^{[t]}(Z) = \zeta^{(1+it)/2}\mathcal{N}^\dagger\big(\mu^{-(1+it)/2} Z \mu^{-(1-it)/2}\big)\zeta^{(1-it)/2}$$

against $\beta_0(t) = \pi/[2(\cosh(\pi t) + 1)]$. Inverse powers act on the respective supports. Cyclicity and the adjoint identity give $\operatorname{Tr} \mathcal{R}^{[t]}(Z) = \operatorname{Tr} Z\Pi_\mu$. The map has a CPTP extension to the whole output space:

$$Z \longmapsto \mathcal{R}_{\zeta,\mathcal{N}}(Z) + \operatorname{Tr}[(I - \Pi_\mu)Z]\,\tau, \tag{3.2}$$

where τ is any fixed input density. The original recovery annihilates off-support blocks, and the additional term is completely positive. This extension is common to every target supported in ζ.

Measured relative entropy obeys

$$D_{M,2}(\alpha\|\beta) \ge -2\log_2 F(\alpha, \beta). \tag{3.3}$$

One may use a fidelity-minimizing Fuchs–Caves measurement [2] and classical Jensen to obtain this inequality. The factor two corresponds to root fidelity.

Proof of Theorem 1.2. Put $\zeta = (\eta_0+\eta_1)/2$, $\sigma_i = \mathcal{N}\eta_i$, and $\mu = (\sigma_0+\sigma_1)/2$. Each target is supported in its corresponding average. Define

$$g_i = D_2(\eta_i\|\zeta) - D_2(\sigma_i\|\mu) \geq 0, \qquad \lambda = \frac{g_0+g_1}{2} = \chi(\eta_0,\eta_1) - \chi(\sigma_0,\sigma_1).$$

Use (3.1) with the same $\zeta, \mathcal{N}$ for both i, and the common completion (3.2). Writing $f_i = F(\eta_i, \mathcal{R}\sigma_i)$,

$$g_i \geq -2\log_2 f_i, \qquad B_i^2 := 2(1-f_i) \leq (\ln 2) g_i.$$

The last inequality follows from $f_i \geq e^{-(\ln 2)g_i/2}$ and $1 - e^{-x} \leq x$. Root fidelity is monotone under a channel, so

$$F(\mathcal{R}\sigma_0, \mathcal{R}\sigma_1) \geq F(\sigma_0,\sigma_1).$$

For any densities a, a', b,

$$|F(a,b) - F(a',b)| \leq \sqrt{2-2F(a,a')}. \tag{3.4}$$

To verify this, choose unit purifications of a, b attaining their root fidelity, and choose a purification of a' aligned with that of a with real overlap $F(a,a')$. The polar-decomposition formula for fidelity permits these choices in a sufficiently large finite purifying space. The difference of the first and third vectors has norm $\sqrt{2-2F(a,a')}$; the inner-product triangle gives one inequality, and swapping a, a' gives the other. The same formula also proves fidelity monotonicity by applying a Stinespring isometry to maximizing purifications. Apply (3.4) in both arguments:

$$0 \leq F(\sigma_0,\sigma_1) - F(\eta_0,\eta_1) \leq B_0 + B_1.$$

Hence the square of the difference is at most

$$2(B_0^2 + B_1^2) \leq 2\ln 2 (g_0+g_1) = 4\ln 2\, \lambda,$$

as asserted. All support and purification arguments allow singular states. □

4 Affinity under a partial trace

Lemma 4.1. *For finite densities ρ_{AB}, σ_{AB},*

$$\operatorname{Tr}\sqrt{\rho_{AB}}\sqrt{\sigma_{AB}} \leq \operatorname{Tr}\sqrt{\rho_A}\sqrt{\sigma_A}. \tag{4.1}$$

Proof. First assume both matrices are strictly positive. On Hilbert–Schmidt matrix spaces let

$$\Delta_{AB}(Z) = \sigma_{AB} Z \rho_{AB}^{-1}, \qquad \Delta_A(Z) = \sigma_A Z \rho_A^{-1}, \qquad W(Z) = (Z\rho_A^{-1/2} \otimes I_B)\sqrt{\rho_{AB}}.$$

Partial trace gives $W^\dagger W = I$, and direct trace expansion gives

$$\langle W(X), \Delta_{AB} W(Y)\rangle_{\mathrm{HS}} = \operatorname{Tr} X^\dagger \sigma_A Y \rho_A^{-1}, \qquad W^\dagger \Delta_{AB} W = \Delta_A.$$

For a positive M, inverse compression satisfies $(W^\dagger M W)^{-1} \leq W^\dagger M^{-1} W$, by the block Schur complement after extending W to a unitary. Integrating this inequality for $M = \Delta_{AB} + sI$ in

$$\sqrt{x} = \frac{1}{\pi}\int_0^\infty s^{-1/2}\frac{x}{x+s}\,ds$$

gives $\sqrt{\Delta_A} \geq W^\dagger \sqrt{\Delta_{AB}} W$. The left and right multiplication operators commute and are positive. Thus $\sqrt{\Delta_{AB}}(Z) = \sqrt{\sigma_{AB}} Z \rho_{AB}^{-1/2}$. Evaluate the compression inequality on $\sqrt{\rho_A}$, whose image under W is $\sqrt{\rho_{AB}}$. The two quadratic forms are exactly the two sides of (4.1).

For semidefinite matrices, replace both by $(1-\varepsilon)\rho + \varepsilon I/d$ on the original fixed space. Their marginals converge to the original marginals. Continuity of square root and trace proves (4.1) in the limit. □

5 Joint convexity of the column deficit

Lemma 5.1. *The function*

$$T(b,u) = \phi(b^2) - \phi(b^2 - u^2)$$

is jointly convex on $\mathcal{D} = \{(b,u) : 0 \le b \le 1,\ |u| \le b\}$.

Proof. Multiply by $\ln 2$ and set

$$f(x) = -\frac{1+x}{2}\ln\frac{1+x}{2} - \frac{1-x}{2}\ln\frac{1-x}{2}, \quad A(q) = \frac{\operatorname{atanh}\sqrt{q}}{\sqrt{q}}, \quad K(q) = \frac{A(q)-1}{q}.$$

Continuous values at zero are $A(0) = 1, K(0) = 1/3$. In the interior put $q = 1 - b^2$, $Q = q + u^2$, and $\delta = u^2$. Then $T_{\mathrm{nat}} = f(\sqrt{q}) - f(\sqrt{Q})$ and $2qA'(q) = (1-q)^{-1} - A(q)$. Differentiation yields

$$\begin{aligned} T_{bb} &= K(q) - A(Q) + 2b^2A'(Q), & T_{bu} &= -2buA'(Q), \\ T_{uu} &= A(Q) + 2u^2A'(Q) > 0, & & \\ \det \nabla^2 T_{\mathrm{nat}} &= K(q)\big[A(Q) + 2\delta A'(Q)\big] - A(Q)K(Q). & & \end{aligned} \tag{5.1}$$

The integral

$$K(q) = \int_0^1 \frac{t^2}{1-qt^2}\,dt$$

and Cauchy–Schwarz give $KK'' \ge 2(K')^2$. Indeed use factors $t/(1-qt^2)^{1/2}$ and $t^3/(1-qt^2)^{3/2}$. Therefore $1/K$ is concave on $[0,1)$.

We also have $2A'K - AK' > 0$. To establish the sign explicitly, expand

$$A(q) = \sum_{n\ge0} \frac{q^n}{2n+1}, \qquad K(q) = \sum_{n\ge0} \frac{q^n}{2n+3}.$$

Let $S_n = \sum_{j=0}^{n} (2j+3)^{-1}$. The coefficient of q^n in $2A'K - AK'$ equals $E_n/[4(n+3)]$, where

$$E_n = 2(n+4)S_n - (2n+3) + \frac{3}{2n+5}, \qquad E_0 = \frac{4}{15}.$$

For verification, the coefficient in $2A'K$ is $(n+2)S_n/(n+3)$, while that in AK' is

$$\frac{1}{2}\sum_{i=0}^{n}\frac{1}{2i+1} - \frac{3}{4(n+3)}\left(\sum_{i=0}^{n}\frac{1}{2i+1} + \sum_{j=0}^{n}\frac{1}{2j+5}\right).$$

Their difference gives the displayed expression. Using $S_n \ge (n+1)/(2n+3)$,

$$E_{n+1} - E_n = 2S_n - \frac{2n}{2n+5} - \frac{6}{(2n+5)(2n+7)} \ge \frac{16n^2+64n+52}{(2n+3)(2n+5)(2n+7)} > 0.$$

Absolute convergence justifies all series products and derivatives.

The tangent inequality for the concave reciprocal, followed by this positive sign, gives for $\delta > 0$

$$\frac{1}{K(q)} - \frac{1}{K(Q)} \le \frac{\delta K'(Q)}{K(Q)^2} < \frac{2\delta A'(Q)}{A(Q)K(Q)}.$$

Multiplication by $K(q)K(Q)A(Q) > 0$ proves that the determinant in (5.1) is positive. At $u = 0$, $T_{bb} = T_{bu} = 0$ and $T_{uu} = A(q) > 0$. The Hessian is consequently positive semidefinite throughout the interior. Continuity extends convexity to the entire closed triangle, including $b = 0, b = 1, |u| = b$. Dividing by $\ln 2$ restores bits. □

6 Canonical flags and the additive pair bound

Proof of Theorem 1.3. First take strictly positive ρ, σ. Put $A = \sqrt{\rho}$, $B = \sqrt{\sigma}$, and

$$R = \rho^{-1/2}(\rho^{1/2}\sigma\rho^{1/2})^{1/2}\rho^{-1/2}.$$

This is the positive solution of $R\rho R = \sigma$. Choose an orthonormal eigenbasis $|j\rangle$ of R, with eigenvalues $t_j > 0$, and define

$$p_j = \|A|j\rangle\|^2, \quad q_j = \|B|j\rangle\|^2 = t_j^2 p_j, \quad |a_j\rangle = \frac{A|j\rangle}{\sqrt{p_j}}, \quad |b_j\rangle = \frac{B|j\rangle}{\sqrt{q_j}},$$

$$z_j = \langle a_j, b_j\rangle, \quad r_j = \frac{p_j + q_j}{2}, \quad \beta_j = \frac{2\sqrt{p_j q_j}}{p_j + q_j}.$$

The exact moment identities are

$$\sum_j r_j = 1, \qquad \sum_j r_j\beta_j = \operatorname{Tr} R\rho = F, \qquad \sum_j r_j\beta_j \operatorname{Re} z_j = \operatorname{Tr} AB = c. \tag{6.1}$$

Here $\operatorname{Tr} R\rho = \operatorname{Tr}\sqrt{\rho^{1/2}\sigma\rho^{1/2}} = F$. This spectral basis is the fidelity-optimal Fuchs–Caves basis [2]; the identities above specify exactly the part required.

Supply an actual orthogonal flag:

$$\eta_0 = \sum_j p_j |a_j\rangle\langle a_j| \otimes |j\rangle\langle j|, \qquad \eta_1 = \sum_j q_j |b_j\rangle\langle b_j| \otimes |j\rangle\langle j|.$$

Discarding it gives ρ, σ. The orthogonal-block formulas give

$$\widehat{F} = F(\eta_0, \eta_1) = \sum_j r_j\beta_j |z_j|, \qquad \widehat{c} = a(\eta_0, \eta_1) = \sum_j r_j\beta_j |z_j|^2.$$

Weighted Cauchy–Schwarz and Lemma 4.1 imply

$$\widehat{F}^2 \le F\widehat{c} \le Fc. \tag{6.2}$$

Let $\widehat{V} = 1 - \chi(\eta_0, \eta_1)$ and $\lambda = \chi(\eta_0, \eta_1) - \chi(\rho, \sigma)$. The entropy loss is additive:

$$V(\rho, \sigma) = \widehat{V} + \lambda. \tag{6.3}$$

With posterior $w_j = p_j/(p_j + q_j)$, the conditional pure-state mixture has determinant $w_j(1-w_j)(1-|z_j|^2)$. The classical flag entropy cancels, giving exactly

$$\widehat{V} = \sum_j r_j\big[\phi(\beta_j^2) - \phi(\beta_j^2(1 - |z_j|^2))\big].$$

The function ϕ is increasing. Thus, with $u_j = \beta_j \operatorname{Re} z_j$, each bracket is at least $T(\beta_j, u_j)$. Signed u_j satisfy $|u_j| \le \beta_j$; Lemma 5.1, Jensen, and (6.1) give

$$\widehat{V} \ge T(F, c) = \phi(F^2) - \phi(F^2 - c^2).$$

Theorem 1.2 applied to these flagged densities and the actual partial trace gives

$$\lambda \ge \frac{(F - \widehat{F})^2}{4\ln 2} \ge \frac{(F - \sqrt{Fc})^2}{4\ln 2}.$$

The second step uses $c \le F$ and (6.2). Combining it with (6.3) proves (1.6).

For a singular pair on dimension d, apply the faithful result to $\rho_\varepsilon = (1-\varepsilon)\rho + \varepsilon I/d$ and $\sigma_\varepsilon = (1-\varepsilon)\sigma + \varepsilon I/d$. For sufficiently small ε the affinity is positive. Entropy, root fidelity, and affinity are continuous at this fixed dimension. Taking $\varepsilon \downarrow 0$ proves the displayed bound. The flags and their bases need not converge because the final expression depends only on these continuous quantities. □

7 The Bell-edge lower bound and asymptotic

Proof of Theorem 1.1. For $0 < v < 1$, differentiation gives

$$\phi'(v) = \frac{\operatorname{atanh}\sqrt{1-v}}{2\ln 2\sqrt{1-v}}.$$

This is positive and decreasing: the series $\operatorname{atanh} s/s = \sum_{n\geq 0} s^{2n}/(2n+1)$ increases with s. Hence ϕ is concave, and for $0 < F \leq 1$

$$\phi(F^2) - \phi(F^2 - c^2) \geq c^2\phi'(F^2), \qquad \phi'(F^2) \geq \frac{\ln(1/F)}{2\ln 2}.$$

For the second inequality set $s = \sqrt{1-F^2}$ and use

$$\operatorname{atanh} s = \frac{1}{2}\ln\frac{(1+s)^2}{1-s^2} \geq \ln(1/F);$$

division by $s \leq 1$ preserves this lower bound. The endpoints follow by continuity.

For any finite pair of affinity $0 < c \leq 1/4$, let $L = \ln(1/c)$. If $F \leq 4c$, Theorem 1.3 gives

$$V \geq \frac{c^2}{2\ln 2}(L - \ln 4).$$

If $F \geq 4c$, then $\sqrt{Fc} \leq F/2$. Put $r = F/c \geq 4$. Keep both terms of the pair bound:

$$V \geq \frac{c^2}{2\ln 2}\left(L - \ln r + \frac{r^2}{8}\right) \geq \frac{c^2}{2\ln 2}L.$$

The final inequality follows because $r^2/8 - \ln r$ increases for $r \geq 4$ and equals $2 - \ln 4 > 0$ at 4. The dimension-independent lower bound passes through Proposition 2.1 to the original infimum.

For the upper bound set $p = (1 - \sqrt{1-c^2})/2$. The commuting pair $\rho = \operatorname{diag}(1-p, p)$, $\sigma = \operatorname{diag}(p, 1-p)$ has affinity $2\sqrt{p(1-p)} = c$, average $I/2$, and cost $h_2(p) = \phi(c^2)$. Its Gram (2.2) is an explicit feasible extension with conditioning dimension two.

Finally $p = c^2/4 + O(c^4)$ and $-p\ln p - (1-p)\ln(1-p) = p\ln(1/p) + p + O(p^2)$. Therefore

$$\phi(c^2) = \frac{c^2}{2\ln 2}L + \frac{c^2}{4\ln 2}(\ln 4 + 1) + O(c^4 L).$$

The lower in (1.4) differs from the leading term by exactly $-c^2$. Since $c^4 L = O(c^2)$, the two asserted remainder bounds follow. They hold for the all-finite infimum even when feasible conditioning dimensions grow with c. □

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